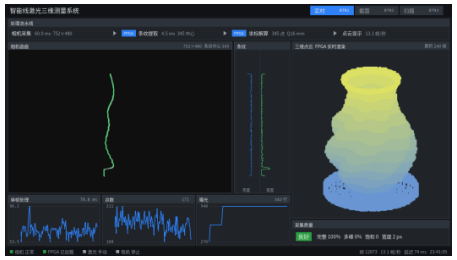
Intelligent Line-Laser 3D Measurement System

INTRODUCTION

Non-contact profile and dimension inspection of small parts. Pixel extraction, 3D geometry and HDMI point-cloud rendering run on one PYNQ-Z2.



BTN1 LIVE

Laser centres
Fixed-point XYZ
Live cloud



BTN2 SECTION

Step heights
Gap indication
Noise and trend



BTN3 SCAN

Angle progress
Cloud accumulation
Result export

Pixel-to-3D hardware pipeline

Single-board operation
752 × 480 grayscale input

→ Data - - -> Control / status

MT9V034 Camera
752 × 480 grayscale

PL / Programmable Logic — 100 MHz

- **Camera Capture / CDC FIFO**
  Frame identity and pixel stream
- **Shared Window / Hessian**
  Shared filter and derivative features
- **Direction Lane 0**
  Direction / refinement
- **Direction Lane 1**
  Direction / refinement
- **Stripe Extraction / Branch Linking**
  Q12 centres and quality flags
- **Fixed-point Geometry Engine**
  Undistort · triangulate · rotate
- **Measure / Section**
  AXI-Lite result registers
- **Voxel Renderer**
  Point buffer / projection
- **DMA / DDR3**
  Raw · PFO3 · XYZ packets
- **Video / HDMI Output**
  VDMA + PL cloud overlay

GP0

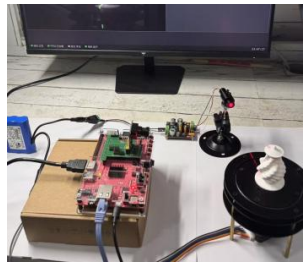
Result readback

PS / ARM Cortex-A9
Scan control, stitching and assessment
HDMI pages, storage and export

Motion Guard / Turntable BTN0 scan · BTN1–3 pages

CREATIVE DESIGN

AMD



On-board test: PYNQ-Z2

RESULTS

PL image processing

26.5 frames/s

Frame-by-frame processing, zero packet errors

Median stripe-centre error **0.02 px**

Synthetic stripes with known centre positions
360° scan: 200 views / ≈2.5 min / ≈66k points

Measurement	Result
Steps, nominal	2 / 3 / 5 mm
Steps, measured	**2.00 / 2.90 / 4.96 mm**
Cylinder diameter error	−0.32 to −0.10 mm
Vase neck error	−0.27 to +0.05 mm

Signed diameter errors against caliper readings.
Cylinder results are calibration rechecks.

Implemented FPGA resources

LUT 73.5% FF 47.8%
BRAM 78.9% DSP 60.9%
WNS +0.230 ns On-chip power ≈2.37 W



Cylinder scan cloud Vase scan mesh

AMD